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HERIOT WATT UNIVERSITY

SCHOOL OF SOCIAL SCIENCES

Edinburgh Business School

C27IE

INTRODUCTORY ECONOMICS

SOLUTIONS

August Examination and Assessment Diet 2024

Duration: Two hours

Section A: Multiple choice questions

There are 20 questions in this section. Please answer all questions at the start of your answer paper. Each question is worth 1.5 marks.

Answer key:

1	**D**	6	**A**	11	**D**	16	**C**
2	**C**	7	**A**	12	**B**	17	**B**
3	**C**	8	**B**	13	**C**	18	**A**
4	**B**	9	**D**	14	**A**	19	**B**
5	**C**	10	**A**	15	**D**	20	**D**

1) The issues that economists study stem from the central economic problem of scarcity. Which of the following is true?

a. Scarcity will eventually disappear with the development of new technology and higher levels of production.
b. Scarcity is an issue for producers, but not for consumers.
c. The opportunity cost principle is important in production, but not for consumers.
d. Whenever there is scarcity, we need to choose which goods and services we desire most.

2) When increasing amounts of a variable factor are used with a given amount of a fixed factor, there will come a point when each extra unit of the variable factor will produce less additional output than the previous unit. We can then say that use of the factor is subject to:

a. Diminishing factor intensity
b. Diminishing marginal cost
c. Diminishing marginal returns
d. Diseconomies of scale

3) At the current market price of 120, there is an excess supply of running shoes. We expect that:

a. The market price will increase, the quantity supplied will increase, and the quantity demanded will decrease.
b. The market price will decrease, the quantity supplied will increase, and the quantity demanded will decrease.

c. The market price will decrease, the quantity supplied will decrease, and the quantity demanded will increase.
d. The market price will increase, the quantity supplied will decrease, and the quantity demanded will increase.

4) Shell produces both petrol and diesel fuels, which are used in different types of engine, so that there is very little possibility of substitution between them (in the short-run). We might expect to find a value of the cross-price elasticity of demand for these products which is:

a. Greater than 1
b. Between 0 and 1
c. Between -1 and 0
d. Less than -1

5) Widespread storms in Brazil have led to a shortage of coffee beans. This can be illustrated by:

a. a leftward shift in the demand curve and a fall in the equilibrium price.
b. a rightward shift in the supply curve and a fall in the equilibrium price.
c. a leftward shift in the supply curve and an increase in the equilibrium price.
d. a rightward shift in the demand curve and an increase in the equilibrium price.

6) Which one of the following would make collusion in an oligopoly more likely?

a. The firms have similar cost structures and production methods.
b. The industry has four firms, who operate separate supply chains
c. Each firm is secretive about costs and production methods
d. The industry has no significant barriers to entry

7) KTM is the main train operator in Malaysia, and so has substantial monopoly power. As a result, we expect that:

a. It will run fewer services than if it faced competition from other businesses.
b. It will not make economic profits in the long-run.
c. It will set prices equal to marginal costs to maximise its profits.
d. It will set lower train fares than if it faced competition from other businesses

8) If we are thinking about production in the long run, which of the following is true?

a. The firm can hire more workers but cannot build a new factory.
b. It is possible that the firm will face decreasing returns to scale.
c. With constant returns to scale, a 1% increase in inputs will lead to a larger percentage increase in output.
d. There will be diminishing marginal returns to some factor.

9) We would expect the demand for a specific brand of coffee beans to be more price elastic than the demand for coffee beans as a product because:

a. Not all consumers are able to grind coffee beans.
b. The market for the brand of coffee beans is much smaller than the market for all coffee beans.
c. Consumers develop a taste for their preferred brand of coffee beans.
d. There is a much higher degree of substitutability among brands of coffee beans than for coffee beans as a product.

10) We define a public good as having the properties of non-excludability and non-rivalry in consumption. Which of the following services comes closest to meeting those requirements

a) Street lighting
b) A golf club.
c) An inter-city coach service
d) A public park

11) After the government imposes a tariff on imports of steel, which group can we be certain will be worse off?

a. Domestic firms which produce steel.
b. The government.
c. Workers employed in the steel industry.
d. Firms which use imported steel in production.

12) Which one of these is a supply side tool for reducing the natural rate of unemployment?

a) Lowering the central bank's interest rate for overnight lending to

commercial banks
b) Abolishing licencing qualifications for skilled workers
c) Decreasing taxes on earned income and employment
d) Increasing the government's capital expenditure

13) Which one of these is an example of an expansionary fiscal policy?

a) Reducing the central bank's interest rates
b) Financing increased government expenditure by increasing the money supply
c) Increasing the income which can be earned without paying income taxes
d) Introducing subsidies for the export of manufactured goods

14) Which of the following is not a basis for calculating the GDP measure of national income?
a) The balance on the capital and current accounts with the rest of the world
b) The expenditure on all final goods and services produced within the economy.
c) The incomes received by households from the provision of factors of production.
d) The value added in the production of goods and services.

15) The real interest rate is

a) the interest rate corrected for deflation
b) the interest rate corrected for unemployment
c) the interest rate corrected for investment
d) the interest rate corrected for inflation

16) Given that the velocity of circulation of money in an economy is 12, real GDP is 1,000bn and the money supply is 95bn. According to the Quantity Theory of Money, the price level will be:

a) 0.86
b) 1
c) 1.14
d) 1.28

17) In which one of these situations might real GDP decrease and the price level remain constant?

a) The aggregate demand and aggregate supply curves both shift right by the same amount.

b) The aggregate demand and aggregate supply curves both shift left by the same amount.
c) The aggregate demand curve does not move and the aggregate supply curve shifts left.
d) The aggregate demand curve shifts left and the aggregate supply curve does not move.

18) Government policy leads to interest rates falling, and both consumption and investment expenditure increasing. With which one of these policies are these effects consistent?

a. A monetary expansion.
b. A fiscal expansion.
c. A fiscal contraction.
d. A monetary contraction.

19) If taxation is 20% of income, there is no saving, and one quarter of disposable income (income after tax has been paid) is spent on imports, then the marginal propensity to spend on domestic goods is:

a. 0.75
b. 0.6
c. 0.55
d. 0.4

20) Which of the following might be expected to shift the aggregate supply curve to the right?

a. The removal of free nursery education for young children.
b. A long period during which interest rates are high.
c. A reduction in government subsidies to firms.
d. The creation of a national apprenticeship scheme.

Section B: Short answer questions

There are four questions in this section, each worth 10 marks. You should answer all four questions.

1) Many countries experienced substantial increases in food prices during 2022.

a) Define the concept of the price elasticity of demand, indicating how it is measured.

(1 marks)

Price elasticity of demand is the responsiveness of the demand for a good to changes in its market price. It is measured as the proportional change in demand, divided by the proportional change in price.

b) If the price of rice has increased from 1.2 per kg to 1.8 per kg, and the demand of a typical household has fallen from 5.5kg per month to 4.5kg per month, calculate the household price elasticity of demand for rise using the mid-point method.

(3 marks)

Proportional change ($\Delta Q/Q$) = 2*(4.5-5.5)/(5.5+4.5) = -0.2;
Proportional change ($\Delta P/P$) = 2*(1.8 – 1.2)/(1.2 + 1.8) = 0.4
PED = ($\Delta Q/Q$)/($\Delta P/P$) = -0.5.

c) State whether the demand for rice, based on the calculation in part b), is elastic or inelastic and explain why this might be the case.

(1 marks)

Since 0 > PED > -1, demand is price inelastic. This is typical of staple foods.

d) Data shows that the overall demand for rice has decreased by 10 per cent over the last five years and this coincides with a period during which real incomes have risen by 15 per cent. Calculate the income elasticity of demand for rice and use the calculation to state whether it is an inferior good or a normal good.

(3 marks)

Income elasticity of demand IED = -2/3. Since IED < 0, rice is an inferior good.

(Question 1 Total: 10 marks)

2. Crystal Clear Audio plans to produce a range of video cameras. It has estimated

how the number of cameras which it could produce in a year will change with the number of workers which it hires.

a) Define the average physical product and the marginal physical product. Complete the table below.
[4 marks]

Average physical product is the number of cameras produced per worker. Divide output by the number of workers.

Marginal physical product is the additional number of cameras which Crystal Clear Audio could produce by hiring one more worker. Divide the change in output from hiring 20 more workers by 20

Workers	0	20	40	60	80	100	120
Output (thousands)	0	60	140	210	270	320	360
APP	-	3,000	3,500	3,500	3,375	3,200	3,000
MPP		3,000	4,000	3,500	3,000	2,500	2,000

b) Sketch a diagram showing the average physical product and the marginal physical product. In your sketch, show clearly where the average physical product and the marginal physical product are equal.
[4 marks]

In a diagram with physical product measured on the vertical axis and the number of workers on the horizontal axis, both curves have maximum points. I expect to see the maximum of the MPP curve between 20 and 40 workers, and the maximum of the APP curve between 40 and 60 workers, with the MPP curve passing through maximum.

c) Assume that Crystal Clear Audio pays its workers 30,000. Given that wages are the only variable costs, complete the table below, showing the average variable cost of output.
[2 marks]

Output (thousands)	0	60	140	210	270	320	360
TVC (thousands)	0	600	1,200	1,800	2,400	3,000	3,600

AVC	-	10	8.57	8.57	8.89	9.37	10

We require TVC = 30,000W, and AVC = TVC/Q

3) Governments can use monetary policy to manage aggregate demand

a. Explain why we treat an increase in interest rates as a form of contractionary monetary policy. Briefly outline how this type of policy can help to reduce the rate of inflation.

(4 marks)

Students should be able to identify two effects of the higher interest rate: a reduction in business's borrowing to fund investment expenditure, and substitution of saving for consumption among households. In AS-AD analysis, both effects cause a reduction in aggregate demand at any price level, which reduce the upward pressure on prices.

b. The UK, and other major economies, have recently suffered from 'stagflation', a combination of high inflation and low economic growth. Explain why the central bank may be torn between adopting contractionary or expansionary monetary policy in these circumstances.

(3 marks)

Most central banks attempt to support price stability and full employment. With low economic growth and high inflation, the bank may need to choose which of these objectives to pursue, since weak growth and employment is usually addressed by an expansionary policy, while high inflation requires a contractionary policy.

c. Explain what would happen to the value of the Euro if the European Central Bank were to set interest at a higher level than in other countries.

(3 marks)

The relatively high rate of interest would encourage saving in Euros. Financial investors would substitute Euro-denominated securities for others. The increased demand would mean that the Euro exchange rate would appreciate.

(Question 3 Total: 10 marks)

4) You are given the following information on an open economy:

Consumer expenditure = 100 + 0.8Y
Investment expenditure = 600
Government expenditure = 1000
Exports = 500
Imports = 0.3Y
There are no taxes

a) Calculate the equilibrium level of income.

(4 marks)

In equilibrium, Y = C + I + G + X – M
So Y = 100 + 0.8Y + 600 + 1,000 + 500 – 0.3Y; and 0.5Y = 2,200, so that Y = 4,400.

b) Suppose that the government believes that the full employment level of income Y^* = 4,800. It proposes to reach that level by encouraging investment.

i. Explain briefly why the increase in national income after an increase in investment is larger than the increase in investment.

(2 marks)

Since higher investment involves the production of goods and services, payments to suppliers of factor inputs, including labour, increase. Household consumption will increase, increasing national income by more than investment.

ii. The multiplier, $m = \Delta Y/\Delta I$, is a measure of the extent to which national income increases relative to the increase in investment. State the relationship between the multiplier and the marginal propensity to withdraw income from the circular flow, and then calculate the value of the investment multiplier in this example.

(3 marks)

Multiplier, m, is the inverse of the marginal propensity to withdraw, MPW, so $m = (MPW)^{-1}$. Here MPW = 0.5, so m = 2.

iii. By how much should investment increase if the economy is to reach its full employment level of national income, Y^* = 4,800?

(1 mark)

The unemployment gap is 400, and the multiplier is 2, so that the gap can be closed by increasing investment by 200.

(Question 4 Total: 10 marks)

Section C – Discussion questions

There are five questions in this section, each of which is worth fifteen marks. You should answer any two questions.

1) Outline the elements of a PEST analysis for a global airline company, such as Emirates Airlines, which is based in Dubai.

Essays will cover political, economics, social and technological elements. It may be easiest to treat these in reverse order.

In terms of technology, students might comment on two important elements of the airline business: the reliance on jet engines, which are responsible for about 2% of all carbon emissions; and booking systems, which through use of surge pricing enable substantial price discrimination in sales.

Turning to social elements, there is the possibility of a link to pressure from popular movements which are seeking a rapid green transition. In addition, students might consider the demand for air travel, noting that this has recovered substantially since the pandemic, and is likely to increase for many years with continuing social and economic development in Asia and Africa.

In economic terms, good answers will note that Emirates' business model relies on being able to provide connections through its hub in Dubai – it is therefore very well suited to transport across Africa, Europe and Asia, using a hub and spoke model – and this gives it a substantial competitive advantage. Referring to the technology used in the industry, some students may note the use of surge pricing to increase revenues on economy seats.

Finally, good answers should note that airlines are often owned by national governments, who provide substantial subsidies. This may lead to higher quality services, but can mean that operating costs are relatively high. The airline industry, operating globally, is also subject to strict safety regulation.

2) When the price of oil increases, the prices of inputs in food production also tend to increase rapidly. An example of this was in 2022, when fertiliser prices increased substantially. In the same year, wholesale grain prices also increased substantially. Discuss how it was possible for farmers to achieve record profits in 2022.

In this case there were price increases in both inputs and final goods. Students should argue that the effect of an input price increase on total costs will depend upon the input's share of total costs. Given the importance of fertiliser as an input, we should expect a substantial increase in production costs. In the market for grain, this should shift the supply curve to the left, so that the quantity traded will be less, while the price will increase. The extent to which the shift in supply is felt on price and quantity will depend on the price elasticity of demand. Where that is high then there will be a reduction in the quantity demanded, and a small increase in prices. In those circumstances, we should expect to see a fall in profits. An increase will follow when demand is sufficiently price inelastic.(Good essays will mention the possibility of substitution in production, reducing costs by changing the mix of inputs.)

3) Applying the structure-conduct-performance approach, what might we conclude about the likely structure of an industry, such as electric power generation in which there are substantial economies of scale?

Essays should begin by stating that in SCP, structure relates to the technological

constraints facing the industry, conduct refers to business behaviour and performance is primarily concerned with business and industry profitability and growth.

We are told that the structure has substantial economies of scale. Essays may include an analysis of natural monopoly, in which firms have a downward sloping average cost curve, with the marginal cost curve lying below it. This means that the structure of the industry will 'naturally' be a monopoly, as the most efficient provider of the service. This should lead to a discussion of profit maximisation in a monopoly, using the condition that marginal cost and revenue will be equal. The producer will constrain output, producing less than if it were to be required to adopt marginal cost pricing, and selling it for a higher price. This enables the firm to make profits in the long-run.

4) For many years, economists have used the term 'business cycle' to refer to the succession of periods of economic expansion, followed by recession, over approximately 7 – 9 years. An important objective of government economic policy is to manage these fluctuations.

a. Briefly outline the links between the business cycle and the credit cycle in banking. [6 marks]

Students should note that the cycle is often associated with changes in the availability of credit, which in turn reflect confidence in the future path of the economy. Banks make investment funds available at times when economic prospects seem positive, and they are confident of borrowers repaying. However, suffering losses from loan failures, banks need to rebuild their balance sheets, and raise more capital. While doing that, they reduce lending, so that investment declines rapidly, and economic growth declines. This can cause a loss of confidence in future economic prospects, leading to a decline in the demand for investment funds.

b. How might governments design their policies so that measures to reduce the fluctuations in the business cycle either occur automatically or else follow rules? [9 marks]

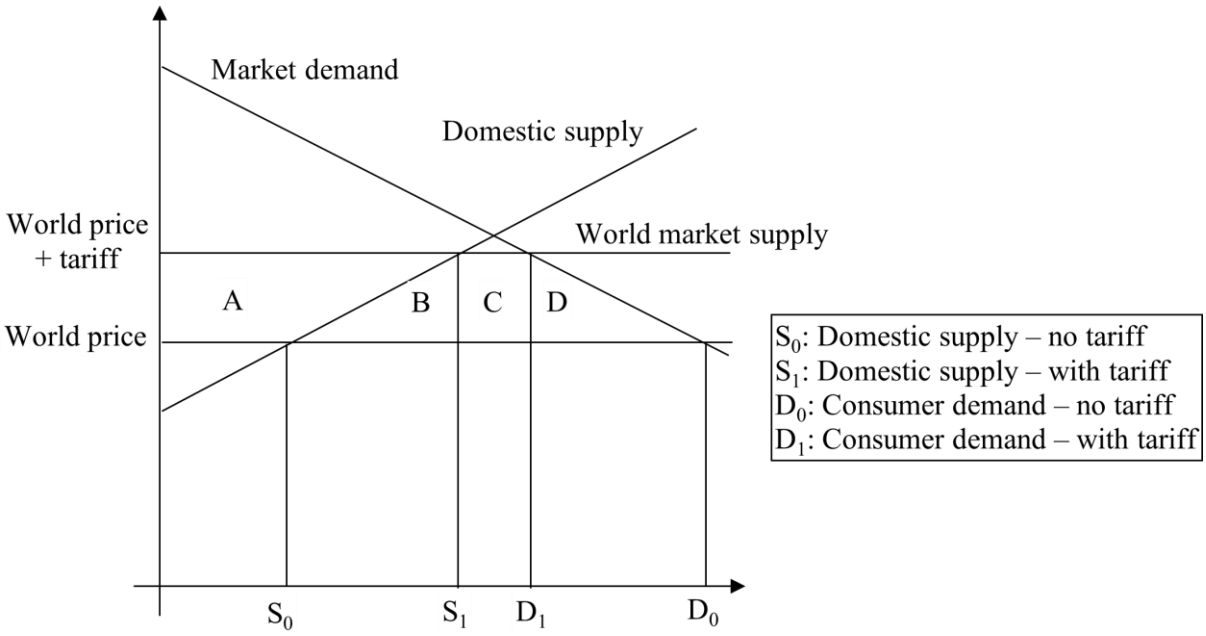
Students should be able to define automatic stabilisers as contingent spending commitments which increase when the economy is weak. Unemployment benefits naturally increase during recessions because of an increase in the number of claims. They should also know that fiscal rules commit the government to varying expenditure as the state of the economy changes.

They should be aware that the alternative to the use of rules is the use of discretion. Giving credit for noting that one reason for democratic governments choosing to follow rules is the risk of increasing volatility though the interaction of the electoral cycle with the business cycle, I expect students to concentrate on the difficulties of designing a discretionary policy – the identification lag, the design and approval lags,

and the implementation lag, so that a policy might only be effective a year or so after the economic conditions occur which justify it, by which time the state of the economy is likely to have changed. Very good answers will note that there is also an argument that the private sector of the economy is more likely to make long-term commitments where the economic environment is predictable, so that discretionary policy may reduce the effectiveness of supply-side policies.

[Total: 15 marks]

5) Outline the economic argument represented in this diagram, to explain how a tariff benefits certain groups in society. Explain briefly why governments may be willing to use tariffs as a policy tool.



Students should explain that the analysis relates to the effects of a tariff in a small country, whose output decision has no effect on global market prices. The diagram shows supply and demand conditions in the domestic market, with competition from international producers. The world market supply curve is perfectly elastic.

Students should argue domestic producers are limited to selling the output which they can produce for less than the international price: S_0 before the tariff and S_1 after. Similarly, total market demand is determined by the world market price: D_0 before the tariff, and D_1 after.

Students should identify that there is a welfare loss to domestic consumers, A+B+C+D. A is acquired by domestic producers, B by suppliers to producers, C as tariff revenue by the government, with D the deadweight loss, indicating that society is worse off after the tariff.

I expect students to argue that within this approach, tariffs can be justified as short-term measures, especially to enable the productivity of an industry to improve, either because it is just being established, or else because it is in need of

restructuring.

END OF PAPER